Discovering Structure

Introduction to NLP — MSc. DH ENC-PSL

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1 Recap

1.1 Language Model



1.1.1 The Language Modeling Problem

Language Model

A Language Model (LM) estimates the probability of pieces of text. Given a sequence of text $w_1, w_2, \cdots, w_S$, it answers the question:

What is $P(w_1, w_2, \cdots, w_S)$?

How to compute P?

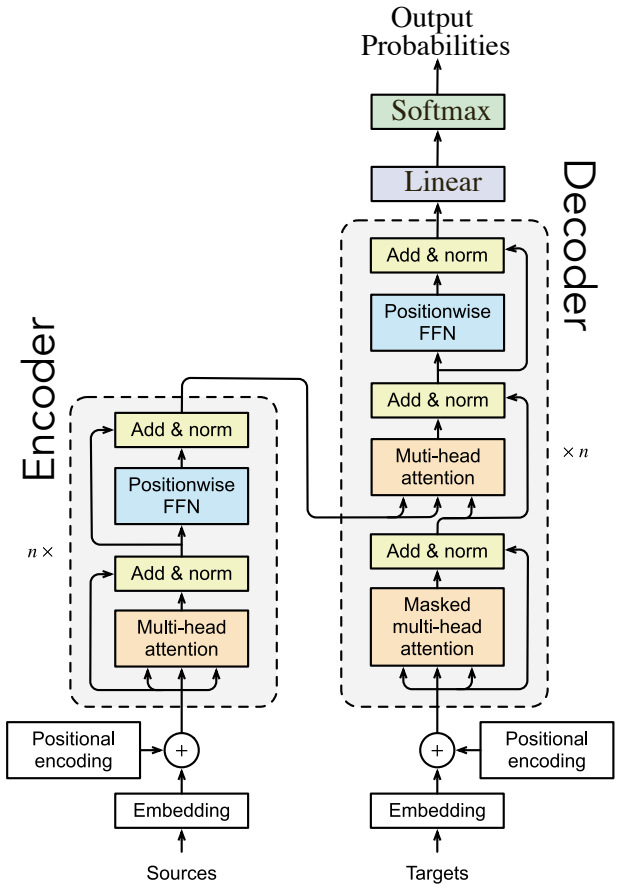
- Count-based (count): n-gram
- Neural method (learn): Transformer
  - Masked Language Modeling (encoder-only, BERT)
  - Causal Language Modeling (decoder-only, GPT)

→ encode:

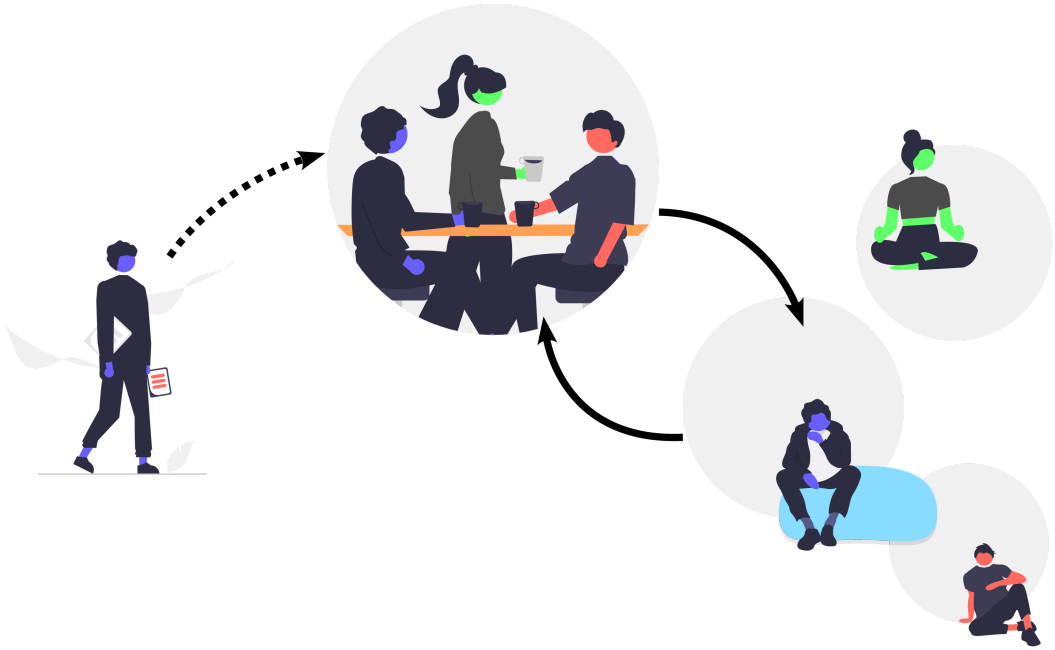
- gramaticality
- semantic plausability
- stylistic consistency
- knowledge (?)

1.2 Transformer

1.2.1 Overview



Transformer Architecture, based on (Vaswani 2017), reworked from (Zhang et al. 2023).



Metaphorical view: *Transformer* = *Support Group*.

1.2.2 Self-Attention: Query-Key-Value

- $\mathbf{Q}$: asking information.
- $\mathbf{K}$: presenting information.

$\rightarrow \mathbf{QK}^{\mathbf{T}}$: *attention weights* from one word to another.

- $\mathbf{V}$: giving information.

$\rightarrow$ information flow scaled by *attention weights*.

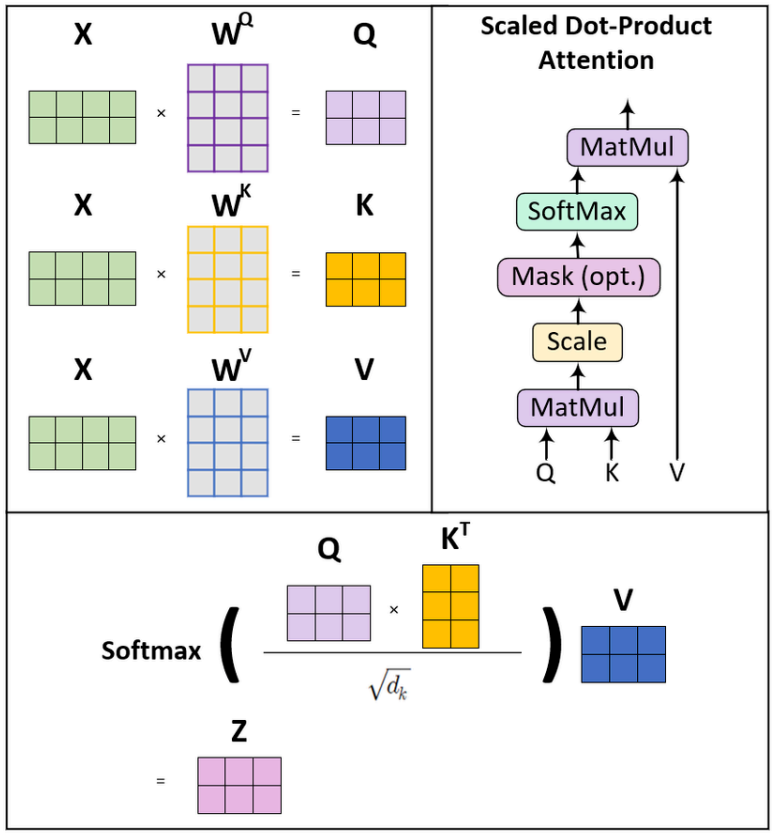


Diagram of the query, key, value, and self-attention mechanism (from (Hwang, Jeong, and Hwang 2025)).

1.3 Pre-trained LM

1.3.1 Word Embeddings — Feature Extraction

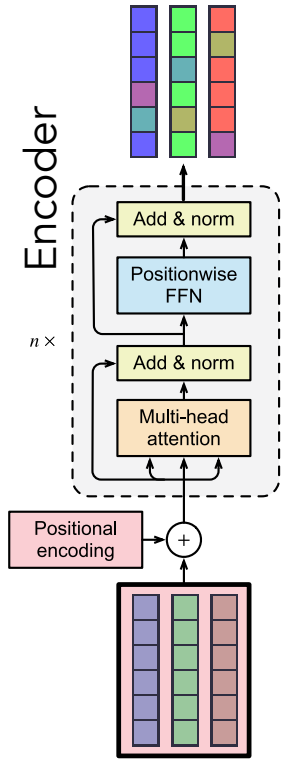


Diagram encoder-only model (e.g.; BERT).

- Tokenize text sequence
  - BERT means Bidirectional Encoder Representations from Transformers.
  - [CLS] bert means bid ##ire ##ction ##al en##code ##r representations from transformers . [SEP]
- Run through the pre-trained LM
  - Output embeddings: $(\mathrm{N}_{\mathrm{batch}} \times)\mathrm{N}_{\mathrm{tokens}} \times \mathrm{d}_{\mathrm{model}}$
- Extract contextual embedding at the poisition of choice

1.3.2 In practice

```python
from transformers import XXModelForTask, XXTokenizer

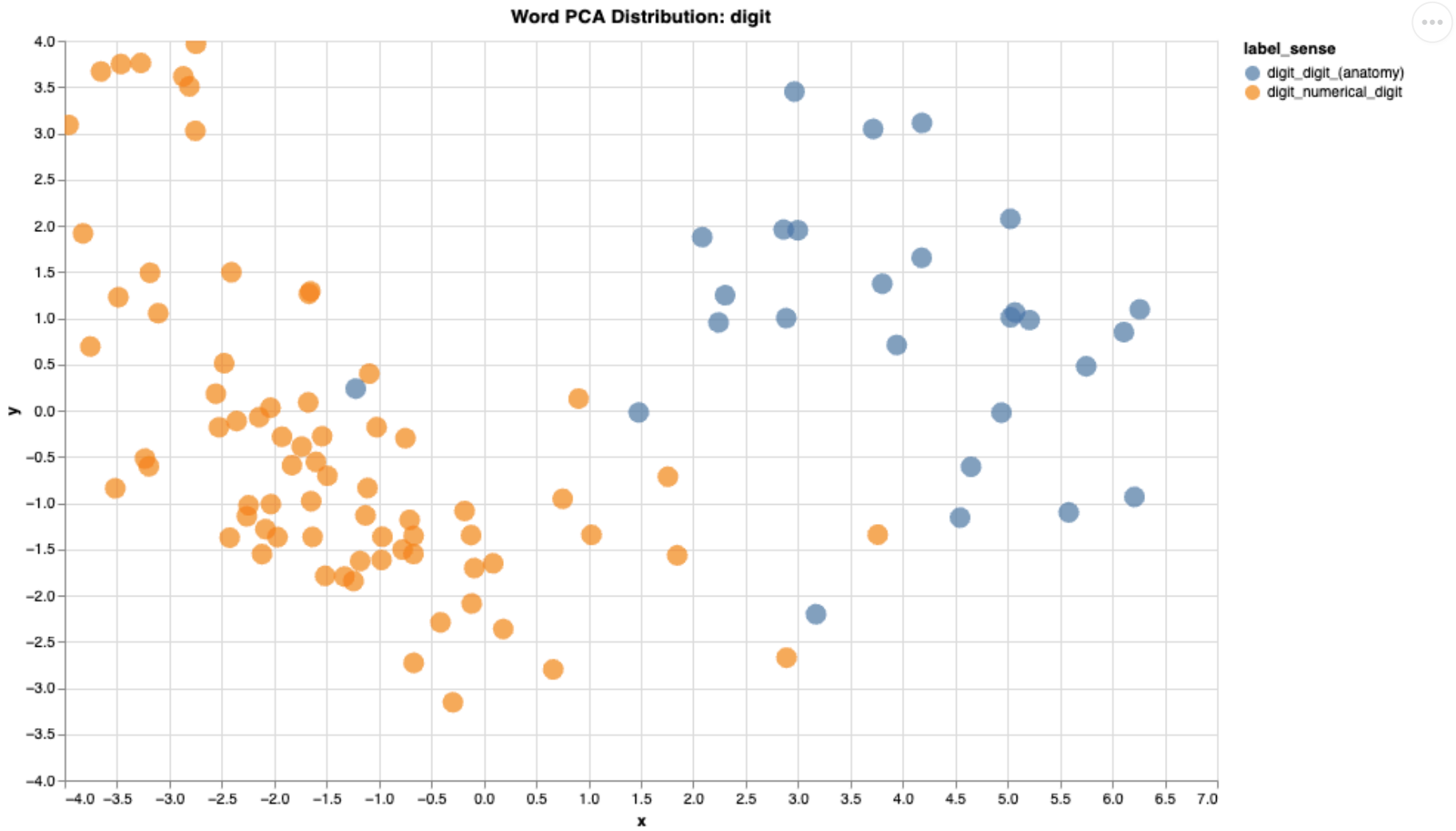
# load the LM
model_name = "<name_of_the_model>"
model = XXModelForTask.from_pretrained(model_name)
tokenizer = XXTokenizer.from_pretrained(model_name)

# load your data
text = "Bla bla bla" # some text you are interested in

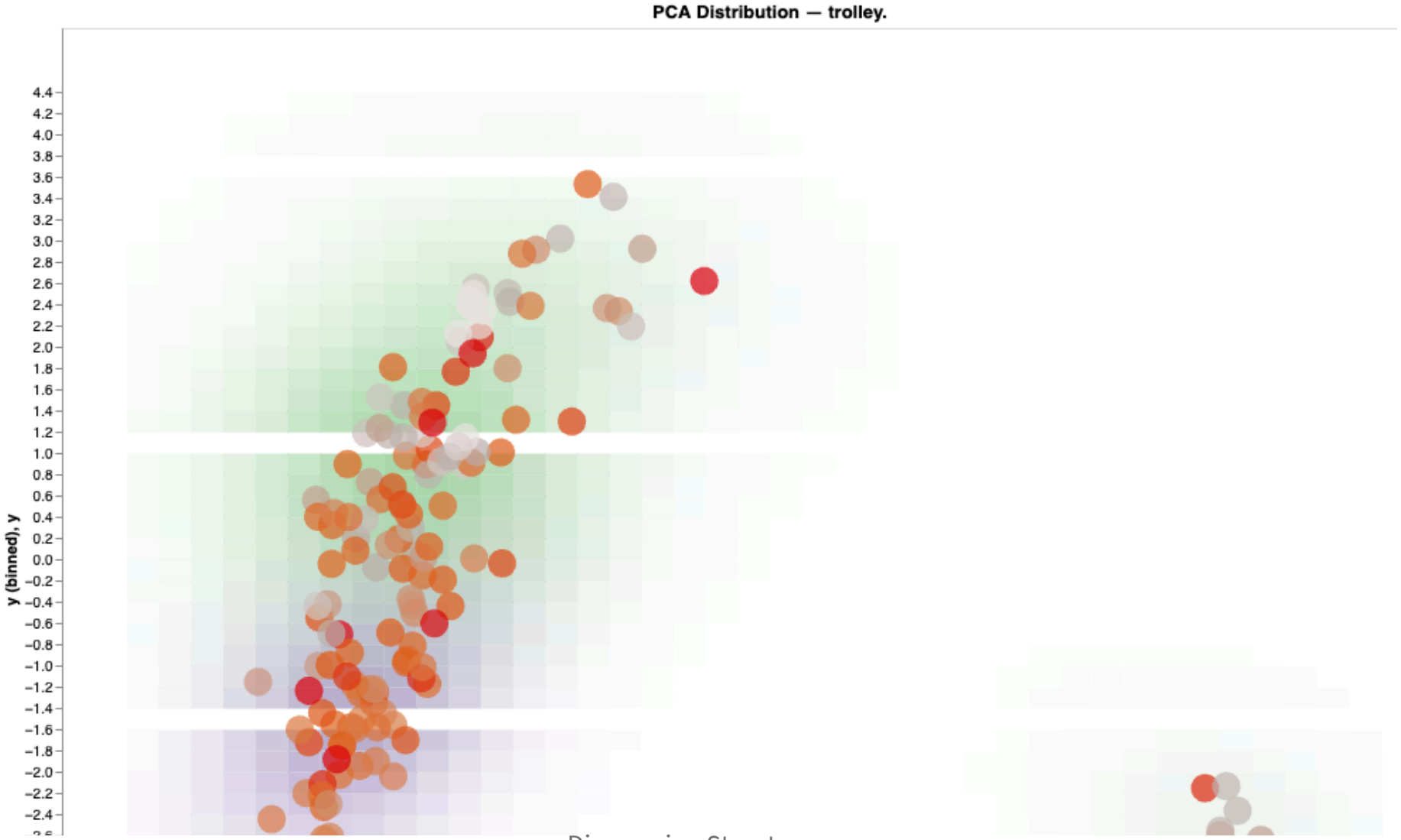
# tokenize the text(s)
tokenized_text = tokenizer(text, return_tensors="pt")

# process it with the model (+ do downstream analysis)
output = model(**tokenized_text)
```

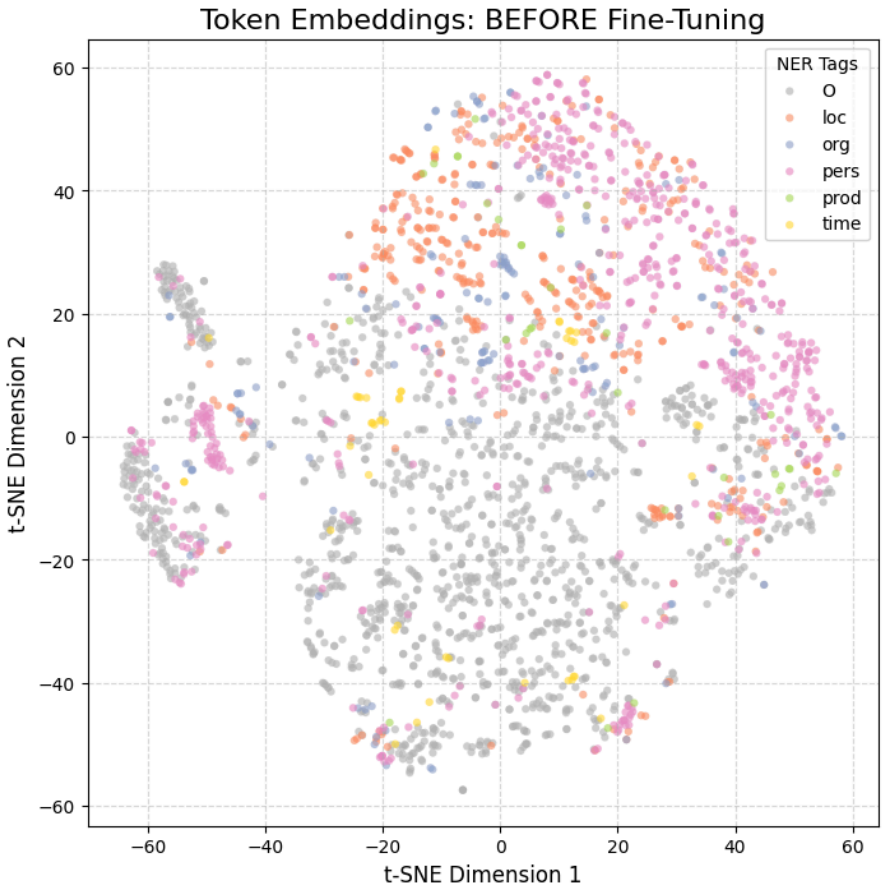
1.3.2.1 Word Sense Disambiguation



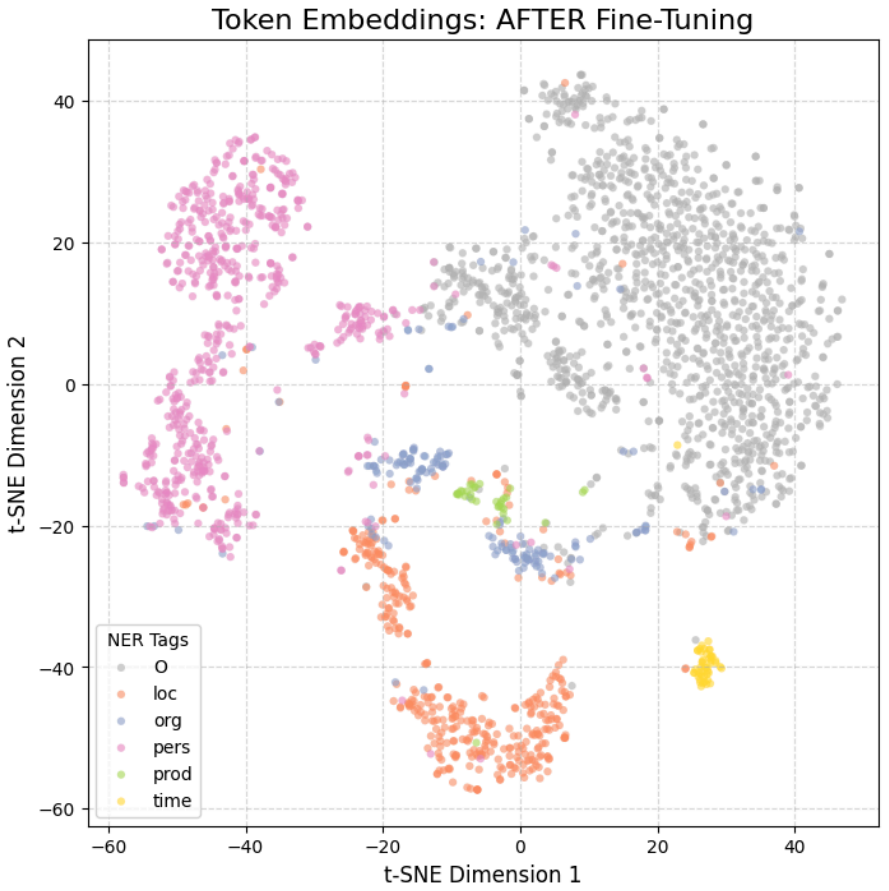
1.3.2.2 Semantic Shift Exploration



1.3.2.3 BERT-based Named Entity Recognition (NER)



t-SNE projections of token embeddings from *pre-trained* `distilcamembert-base`, for 500 samples of the CLEF-HIPE (fr) 'dev' split, colored by NER tag (BI-scheme removed, 'O' tokens sumbsampled).



t-SNE projections of token embeddings from *fine-tuned (5 epochs)* `distilcamembert-base`, for 500 samples of the CLEF-HIPE (fr) 'dev' split, colored by NER tag (BI-scheme removed, 'O' tokens sumbsampled).

Table of Contents

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- 4 Topic Modeling
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2 From Words to Document Representations

2.1 Introduction

2.1.1 Definitions

Representation / Embedding

Numeric vector representation that a machine can understand.

Symbolic Language → LM → Representation



Document = sequence of words (/tokens)

– depending on context and applications (sentence, whole document (short/long), chunks, ...)

2.2 Count-based Approach

2.2.1 Bag-of-Words (BoW)

BoW Definition

Document = **multiset** of the **words** it contains.

Given a vocabulary □:

- BoW represents documents as *vectors* of dimensionality: |□|
- Each component is associated with a token
  - Main variants: binary, **raw count**, (frequency)



BoW Illustation of the first chapter of L'Etranger (Camus, 1942); top 150 words.

2.2.2 TF-IDF

Highlight Relative Importance

Weight the terms according to their specificity across documents in the corpus, based on *term specificity* (Sparck Jones 1972).

$$\text{TFIDF}(t, d, D) = f_{t,d} \cdot \log\left(\frac{N}{n_t}\right)$$

- Term Frequency: $f_{t,d}$
- Inverse Document Frequency: $\log\left(\frac{N}{n_t}\right)$,
  - for a term/token t, document d in corpus D
  - with $N = |D|$: number of documents in the corpus; $n_t = |\{\tilde{d} \in D | t \in \tilde{d}\}|$: number of documents containing term t.

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.966
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

Document Frequency (df) and Inverse Document Frequency (idf) for some words in Shakespeare's 37 plays. Extracted from wikipedia.

2.2.3 Limitations

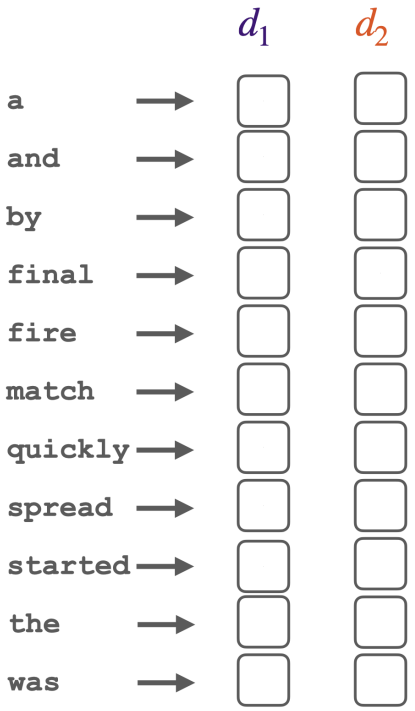
- Context-insensitive
  - No word order (*"The pig and I ate a salad"* = *"I ate the pig and a salad"*)
  - Cannot handle polysemy (*"The match was fire, it was lit!"* ~ *"The fire was lit by a match"*)
- High-dimensional sparse representation
  - Vocabulary can rapidly grow (easily $|\mathrm{V}| > 10^{5+}$) (cf. *curse of dimensionality*)
  - Orthogonality — no concept of semantic similarity at the word-level (e.g.; *"The cat sat on the mat"* ≠ *"A feline rested on the carpet"*)

2.2.4 Quick Example

Find the binary and absolute frequencies BoW and TF-IDF representations for:

- d_1: The final match was fire.
- d_2: The fire was started by a match and the fire spread quickly.

Voc. | Binary BoW | Raw Count BoW | TF-IDF



2.3 Transformer-based



2.3.1 Motivation: Documents comparison

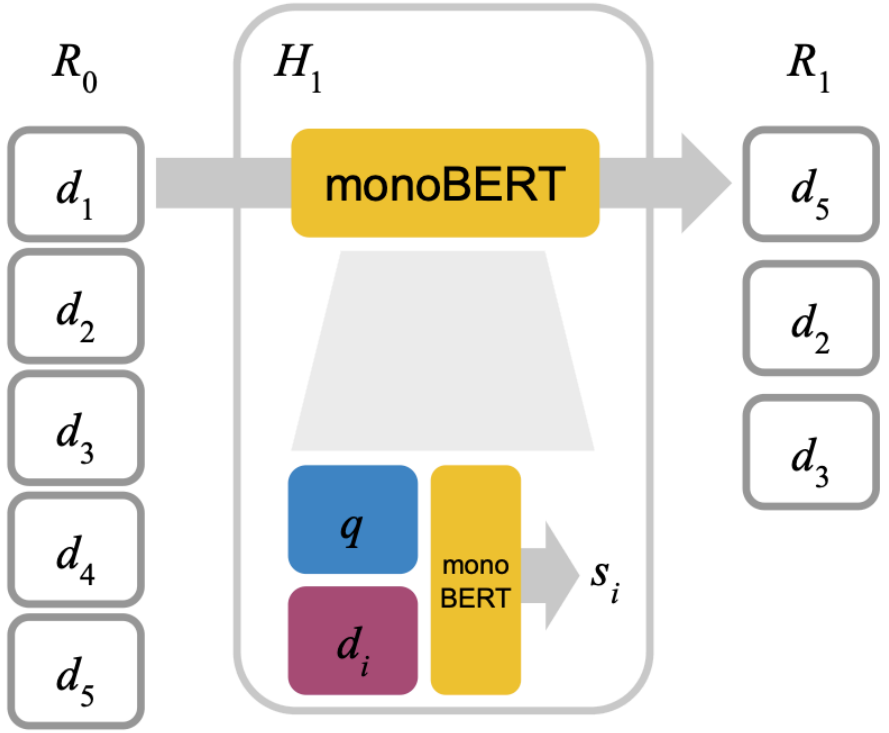
How to compare documents?

→ *pairwise scores through cross-encoder mechanism!*

Computational Bottleneck

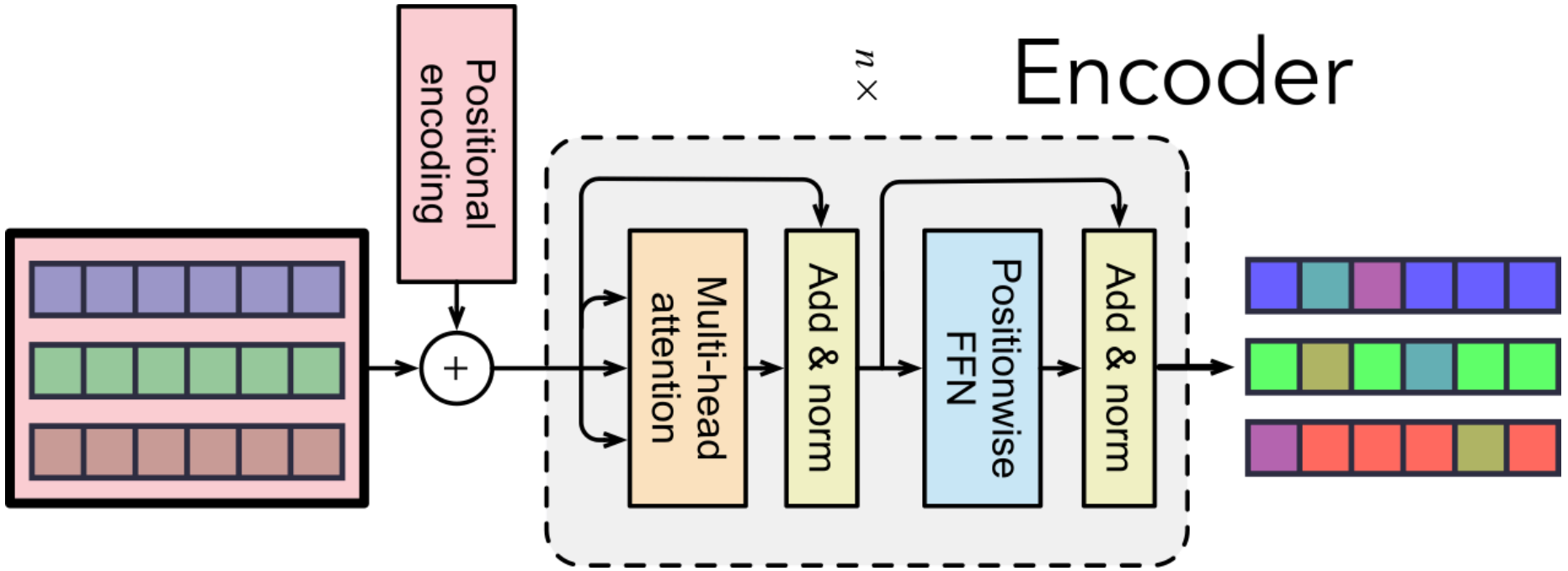
Suppose you have $n = 10\,000$ sentences, how long would it take?

- $n(n-1)/2 = 49\,995\,000$ inference computations
- ≈ 65 hours on a V100 GPU (Reimers and Gurevych 2019)



Cross-encoder logic. *Extracted from (Nogueira et al. 2019) (cropped).*

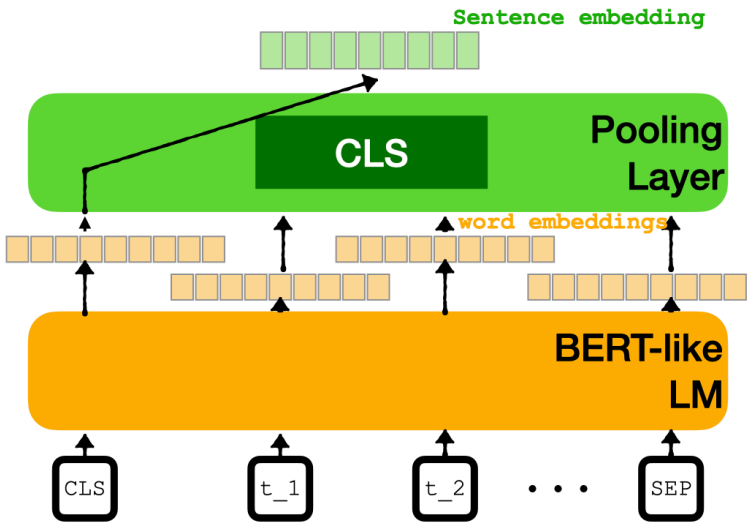
2.3.2 Reminder: Contextual Word Embedding

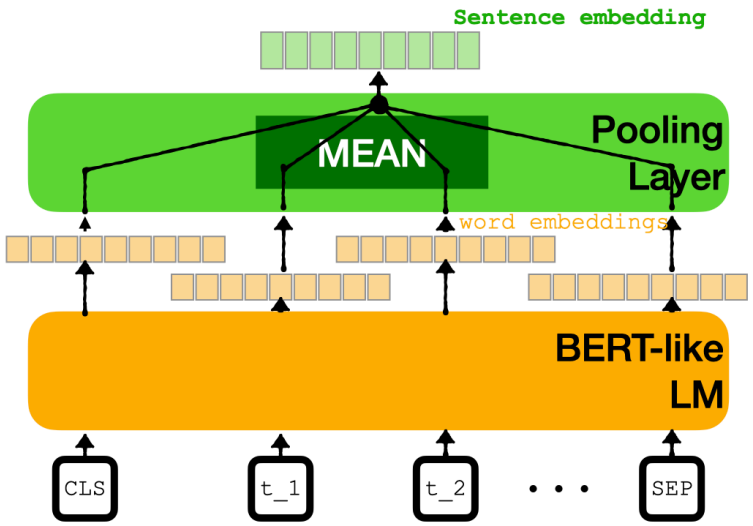


Encoder-based feature extractor diagram (token-level).

→ *How to get representations for a whole document?*

2.3.3 *Naive* Pooling Strategies





→ works to some extent but results are not that good, & unfit for large-scale semantic comparison…

BERT was not trained for sentence-level representation!

2.3.4 SentenceBERT (Reimers and Gurevych 2019)

1908.10084 1 / 11 87%

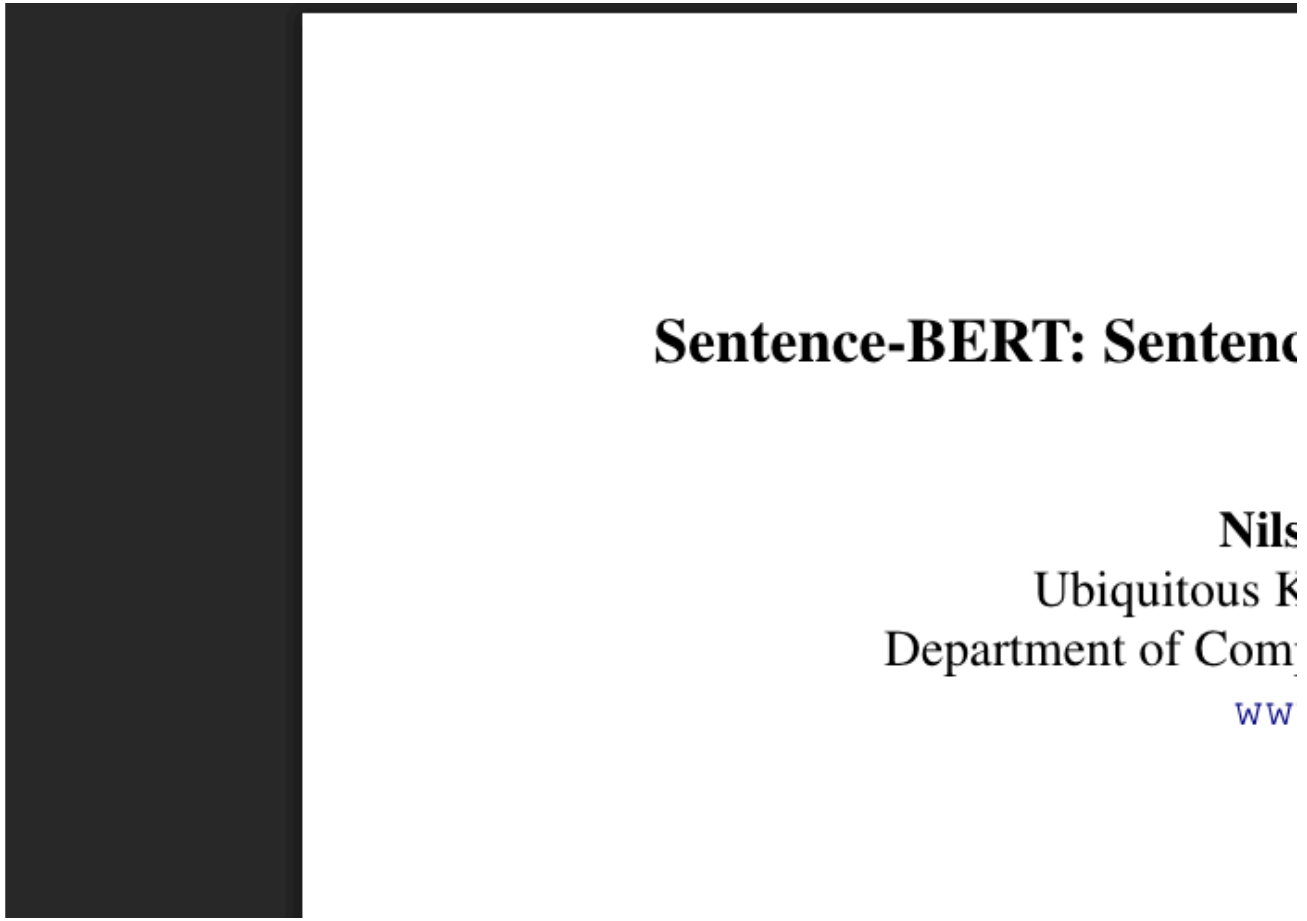


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2.3.5 SBERT="BERT-like model +pooling"

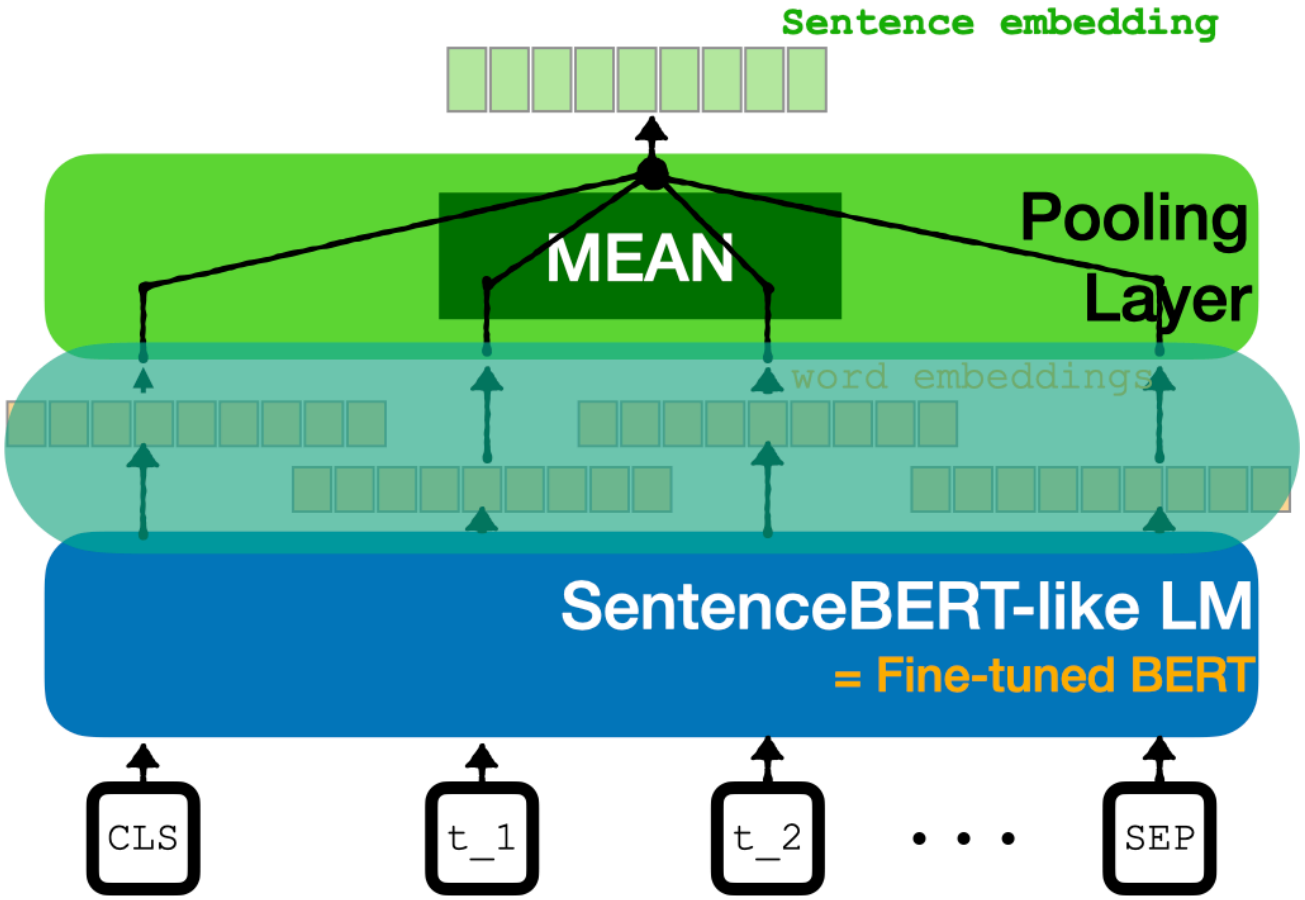
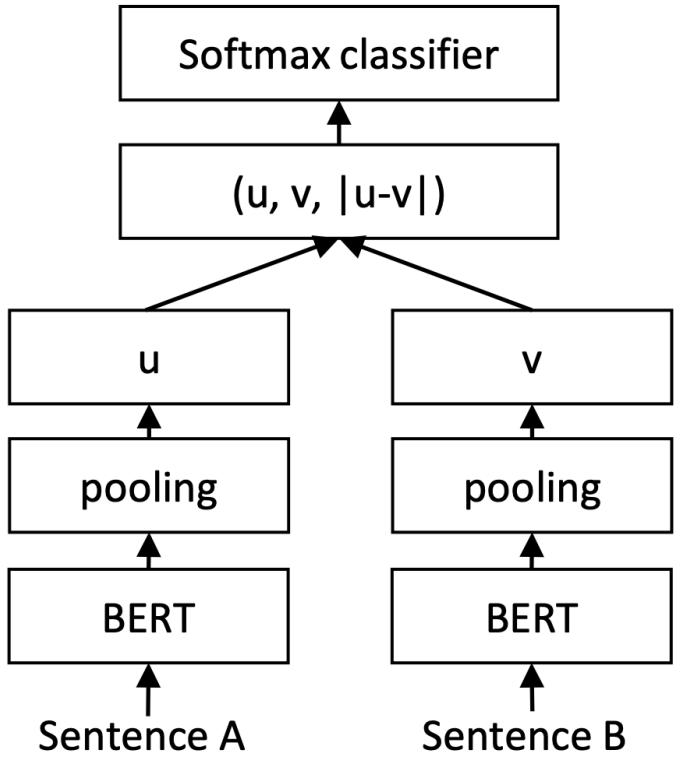


*Illustration of **SentenceBERT**-like models architecture at inference time.*

2.3.6 Training Principle



SBERT architecture with classification objective function. Fig. 1 extracted from (Reimers and Gurevych 2019).

- Siamese network
  - Shared weights
- Aggregation of pooled embeddings
  - CLS, **MEAN**, MAX
- Three objective functions (sequence-level)
  - Classification (**NLI** used to **train** the model)
  - Regression
  - Triplet

2.4 Summary

2.4.1 Key differences

	TF-IDF	SentenceBERT
Approach	Count-based	Neural
Vector	Sparse (d~$10^{5\text{-}6}$)	Dense (d~$10^{2\text{-}3}$)
Interpretabilty	✅	❌
Corpus-dependent	Yes (relative)	No (absolute)
Context	❌	✅
Semantic	❌	✅

3 Document-level Semantic Spaces

3.1 Reminders

3.1.1 Cosine similarity

Dot Product:

$$\mathbf{A} \cdot \mathbf{B} = ||\mathbf{A}||\ ||\mathbf{B}|| \cos \theta$$

Cosine similarity:

$$\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\ \|\mathbf{B}\|}$$

Origins

Use of cosine similarity for related tasks is inherited from Information Retrieval and Vector Space Models (Salton, Wong, and Yang 1975).

No Free Lunch

Mathematically also suffers from the curse of dimensionality, but **empirically** works better in practice (most often)…

… still has know limitations, e.g.:

- can sometimes yield arbitrary (and therefore meaningless) 'similarities' (Steck, Ekanadham, and Kallus 2024);
- relative to human judgements, cosine underestimates similarity of frequent words (Zhou et al. 2022).

3.2 Dimensionality Reduction

3.2.1 From high- to lower-dimensional spaces

Can serve different purposes:

- Mitigate curse of dimensionality
- Visualisation & Exploration
- Analytical

→ various methods exist (MDS, PCA, tSNE, UMAP, …)

3.2.2 Quick Cheatsheet

MDS (Multidimensional Scaling)

- Preserves pairwise **distances**.
- Good for small datasets, metric interpretability.
- Computationally expensive (to untractable) for large datasets.

Key params: `metric`, `n_init`, `max_iter`

PCA (Principal Component Analysis)

- Linear projection maximizing **variance**.
- Fast, deterministic, interpretable axes.
- Captures global but not nonlinear structure.

t-SNE (t-Distributed Stochastic Neighbor Embedding)

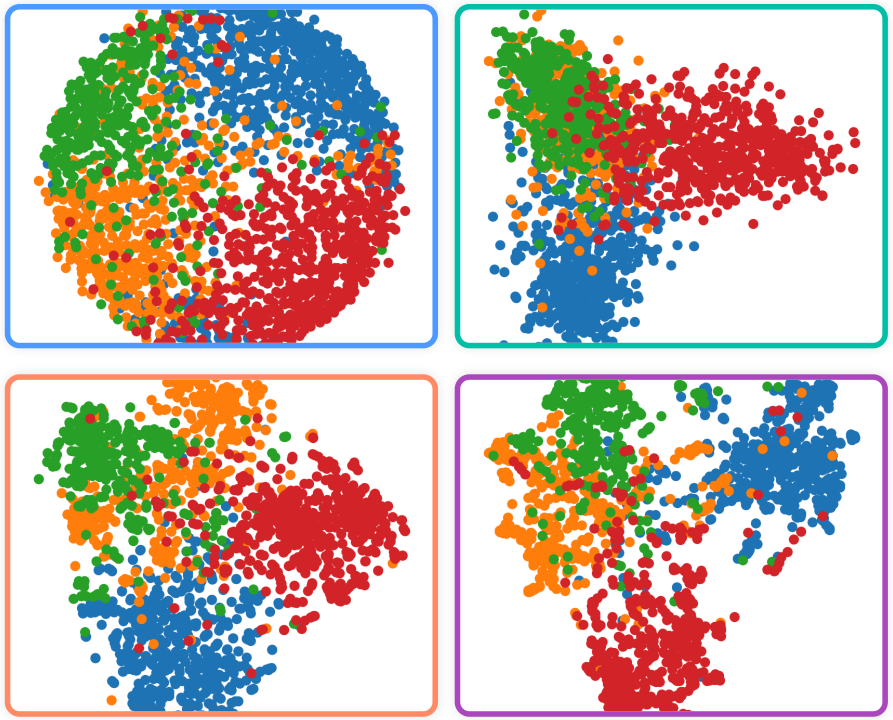
- Preserves **local neighborhoods**.
- Excellent for cluster visualization.
- Nonlinear, stochastic, slower.

Key params: `perplexity`, `learning_rate`, `n_iter`

UMAP (Uniform Manifold Approximation and Projection)

- Preserves **local + global** structure.
- Scalable and fast.
- Good for semantic visualization of embeddings.

Key params: `n_neighbors`, `min_dist`, `metric`

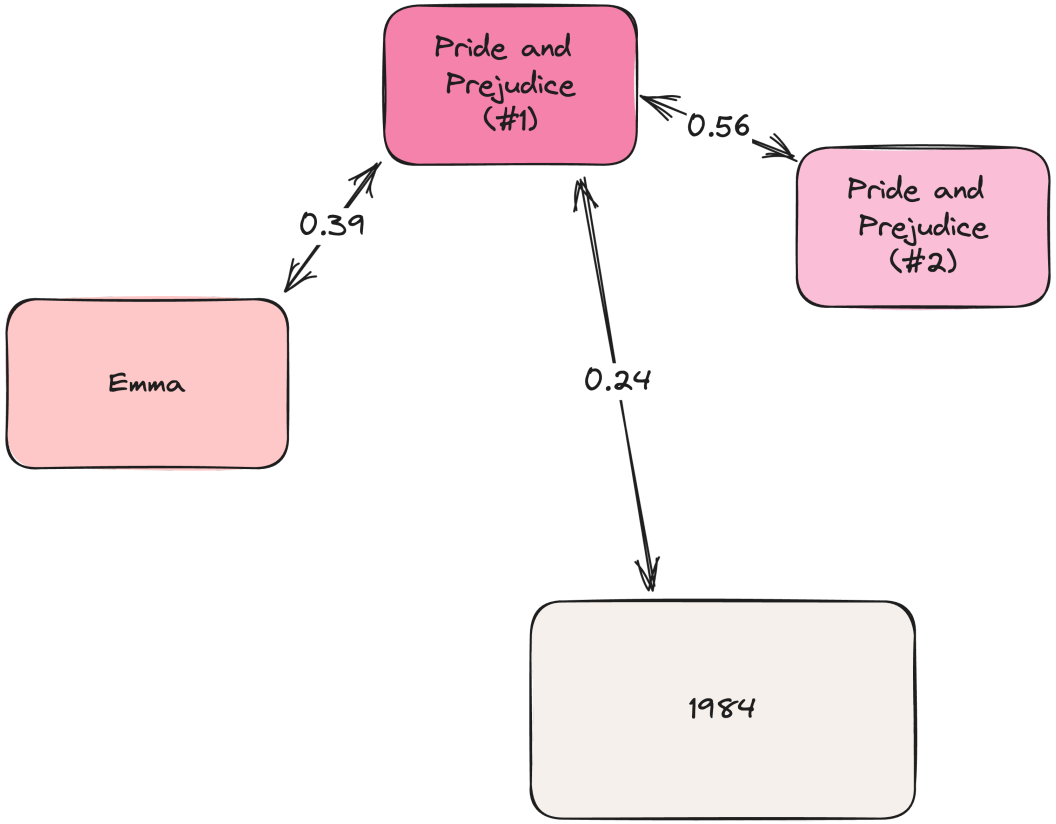


🔬 **Experiment:** Illustrations based on ~2K excerpts of 20Newsgroups, covering 4 topics. Dimensionality reduction methods applied to 384-dimensional embeddings obtained with `all-MiniLM-L6-v2` SentenceTransformer, using cosine distance for MDS and default arguments from `sklearn` and `umap` libraries for PCA, t-SNE, and UMAP.

💡 **Tip:** Use *PCA* for preprocessing and transformation interpretability, *t-SNE* or *UMAP* for visualization, *MDS* for preserving metric geometry.

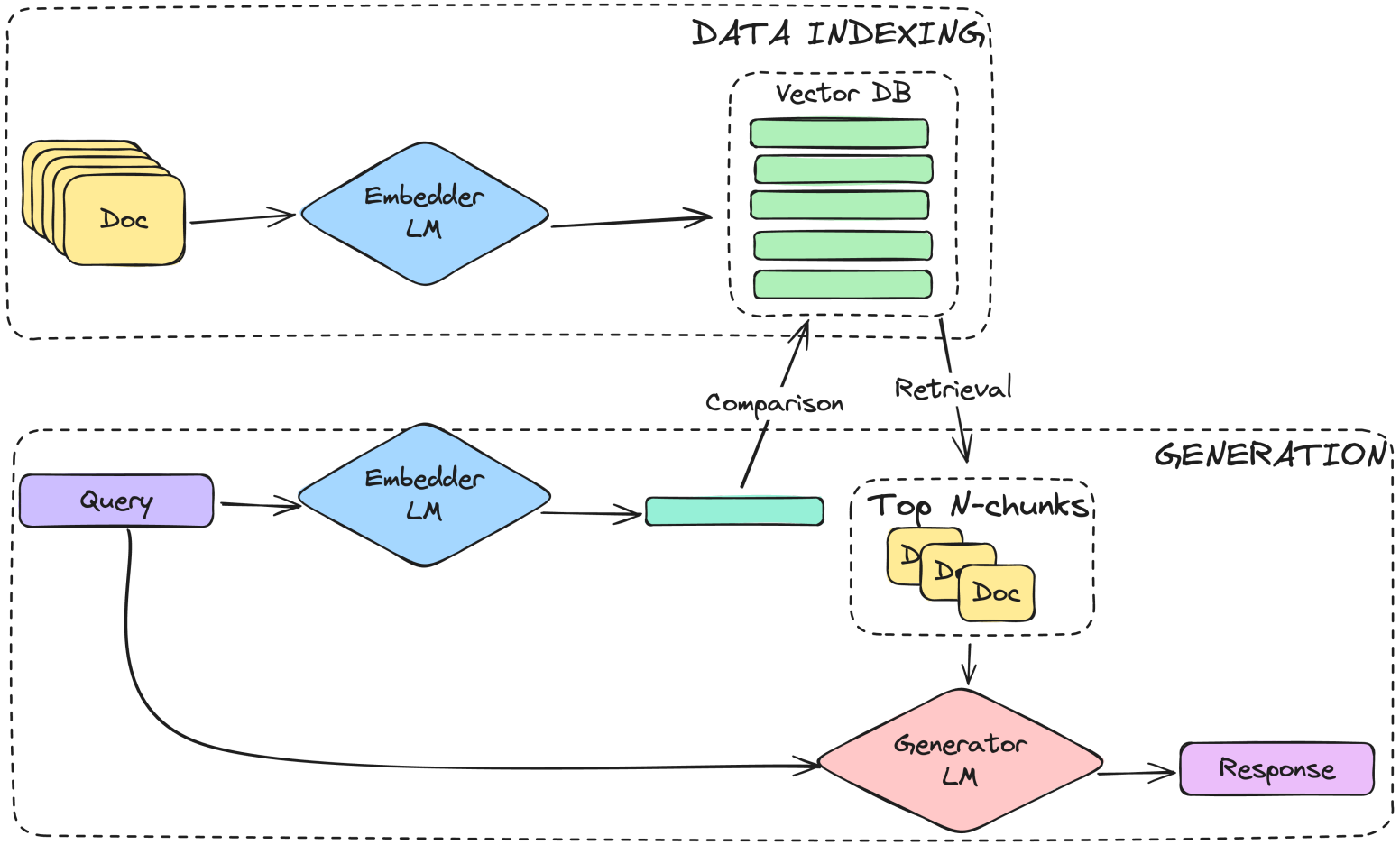
3.3 What can we do with *document distances*?

3.3.1 Compare Documents!



Quick experiment with random chunks from *Pride and Prejudice*, and *Emma* by J. Austen, and *1984* by G. Orwell. Cosine distance computed on embeddings obtained with `all-MiniLM-L6-v2` SentenceTransformer.

3.3.2 Retrieve Documents



Naive Retrieval Architecture Illustration (→ Applied to RAG.)

3.3.3 Group Documents

Overview of common clustering algorithms and their main features.

K-means

- Partitions data into **K clusters** minimizing within-cluster variance.
- Assumes spherical, equally sized clusters.
- Fast, scalable, but sensitive to outliers and initialization.

Key params: `n_clusters`, `init`, `n_init`, `max_iter`

DBSCAN

- Density-based: forms clusters of **closely packed points**.
- Identifies outliers as noise; no need to predefine K.
- Can struggle with varying densities or high-dimensional data.

Key params: `eps`, `min_samples`, `metric`

HDBSCAN

- Hierarchical extension of DBSCAN with **adaptive density**.
- Automatically finds stable clusters and handles noise robustly.
- Excellent for high-dimensional embeddings (e.g., SBERT, UMAP).

Key params: `min_cluster_size`, `min_samples`, `cluster_selection_epsilon`

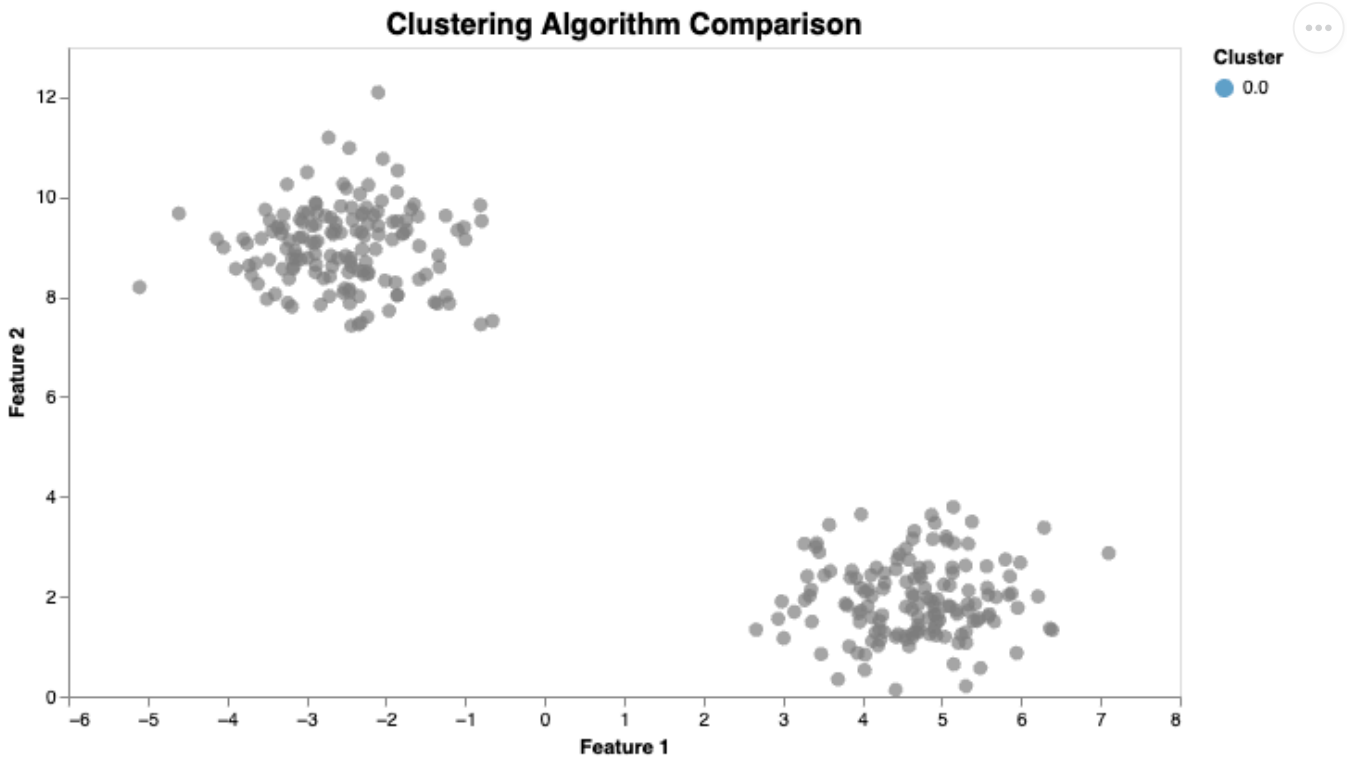
BIRCH

- Incremental clustering using a feature **tree structure**.
- Handles large or streaming datasets efficiently.
- Works best for roughly spherical clusters.

Key params: `threshold`, `branching_factor`, `n_clusters`

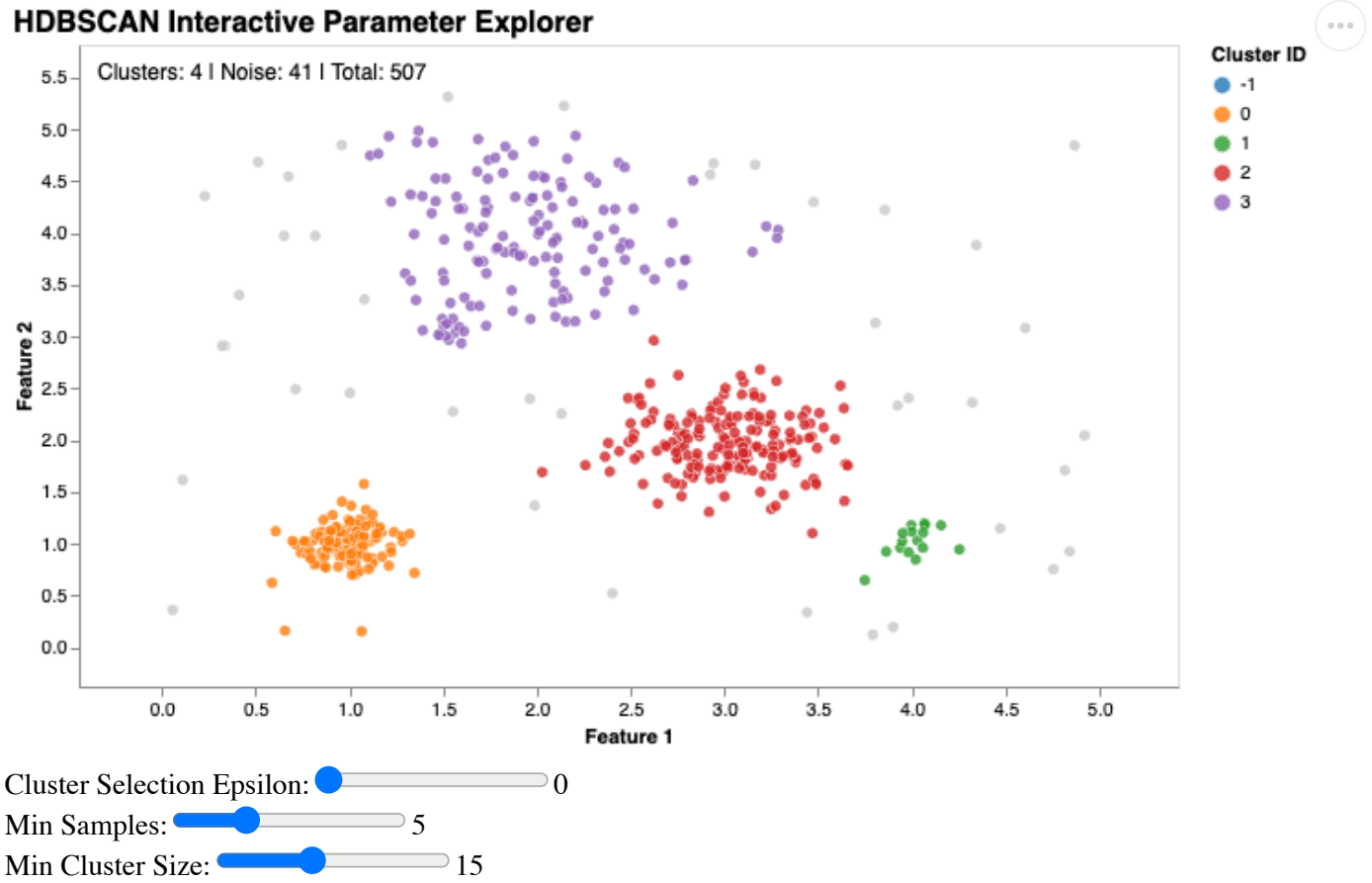
💡 **Tip:** Use *K-means* for simple clusters, *DBSCAN/HDBSCAN* for arbitrary shapes and noise, *BIRCH* for large-scale or online data.

3.3.3.1 Examples



Dataset: Simple: Two Clusters
Clustering Method: None

3.3.3.2 HDBSCAN Focus



4 Topic Modeling

4.1 Introduction

4.1.1 Motivation

UN General Debates (1946-present)

- Decades of speeches by world leaders
- Millions of words across thousands of transcripts

→ What are world leaders *really* talking about?

UNGDC Topic Modeling Hands-on

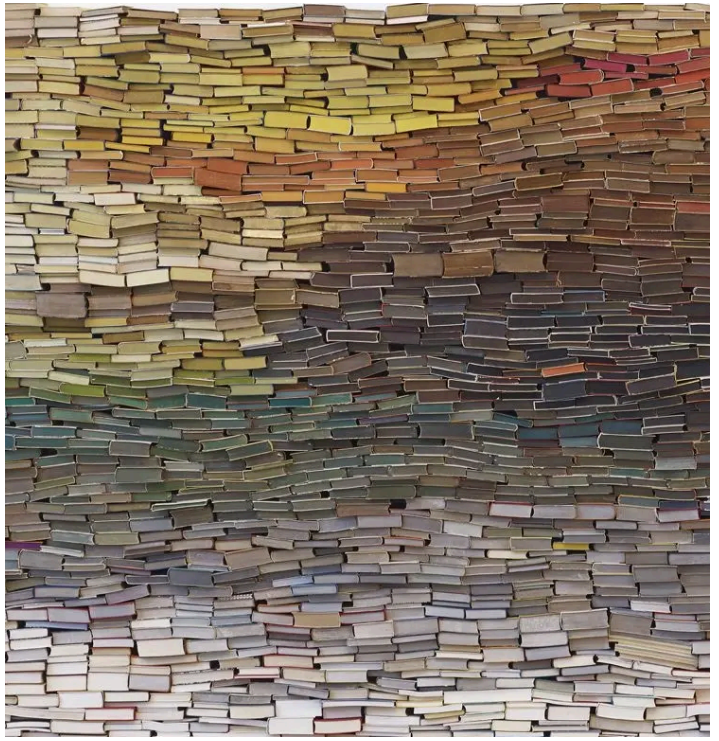
This will be your task for today's Hands-on session.



UN General Assembly Hall, NY headquarters. ©Basil D Soufi (via WikiCommons), 2011.

4.1.2 Description

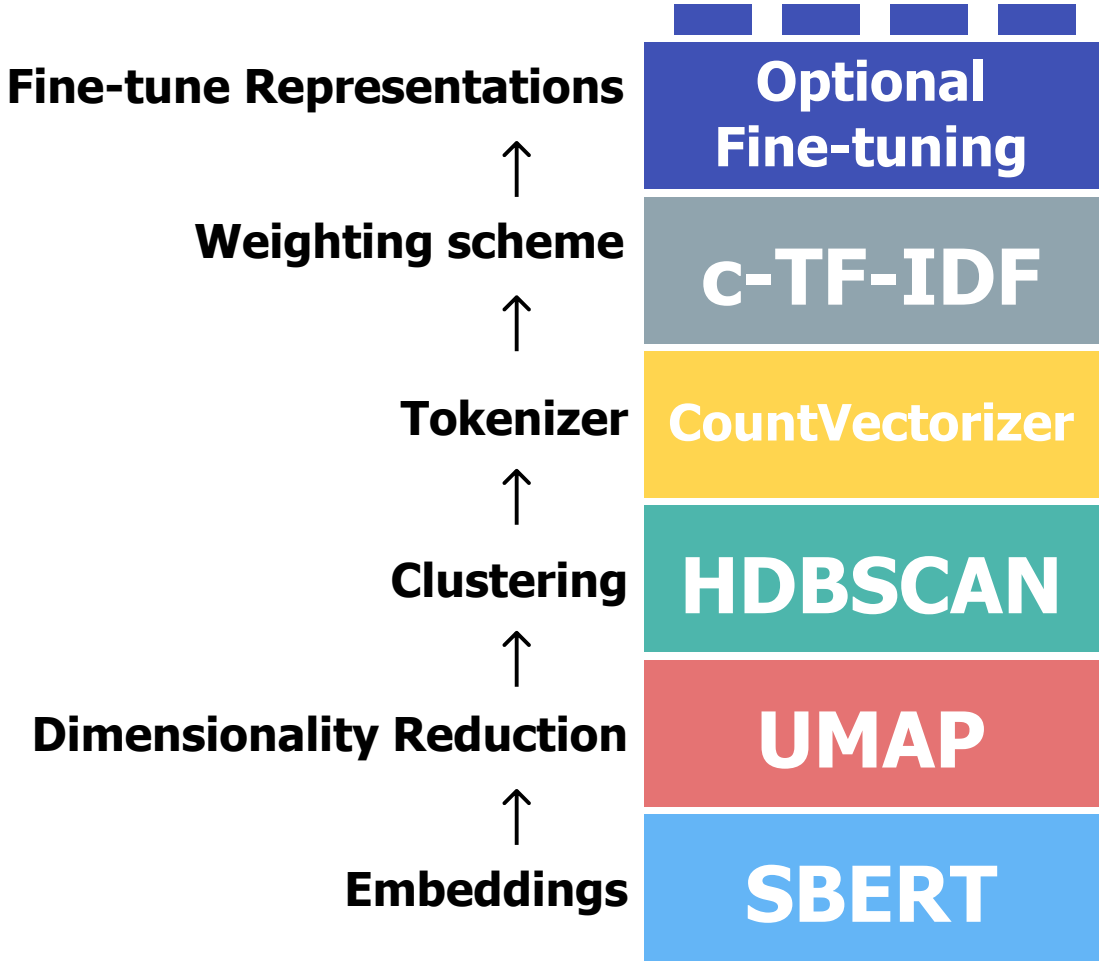
- Topic Model: *unsupervised* algorithm
  - Discover main ‘themes’ in an unstructured corpus
- Multitude of methods
  - “proper” modeling
  - adapted tasks



Enclosed Content Chatting Away In The Colour Invisibilty, 2009, Anouk Kruithof. (cropped).

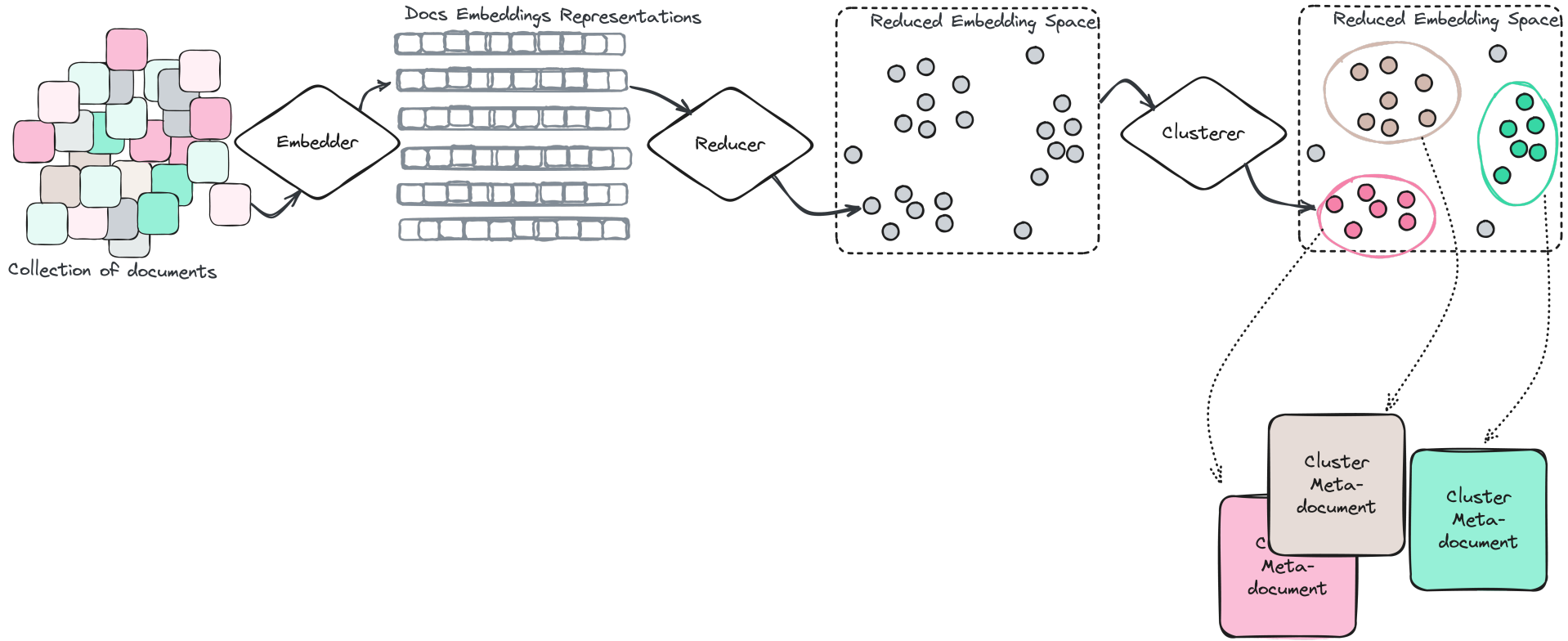
4.2 BERTopic

4.2.1 BERTopic Overview

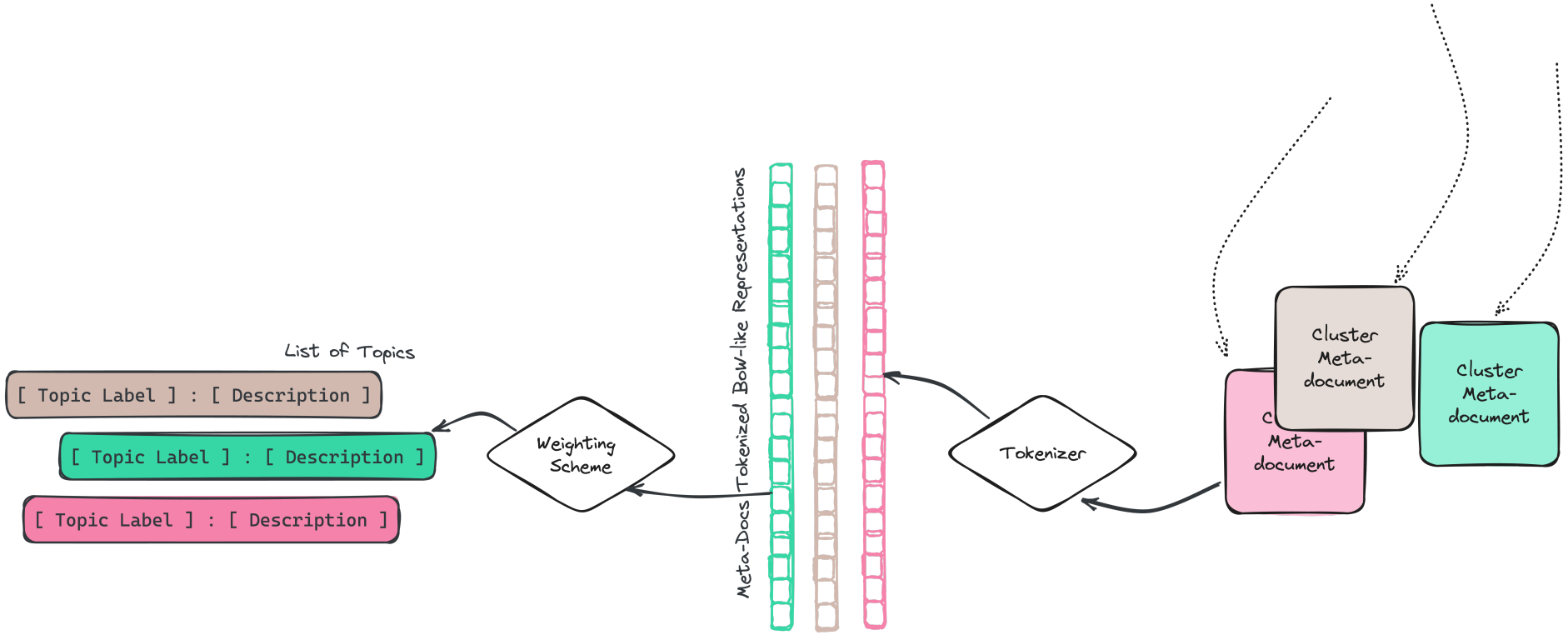


From (Grootendorst 2022)

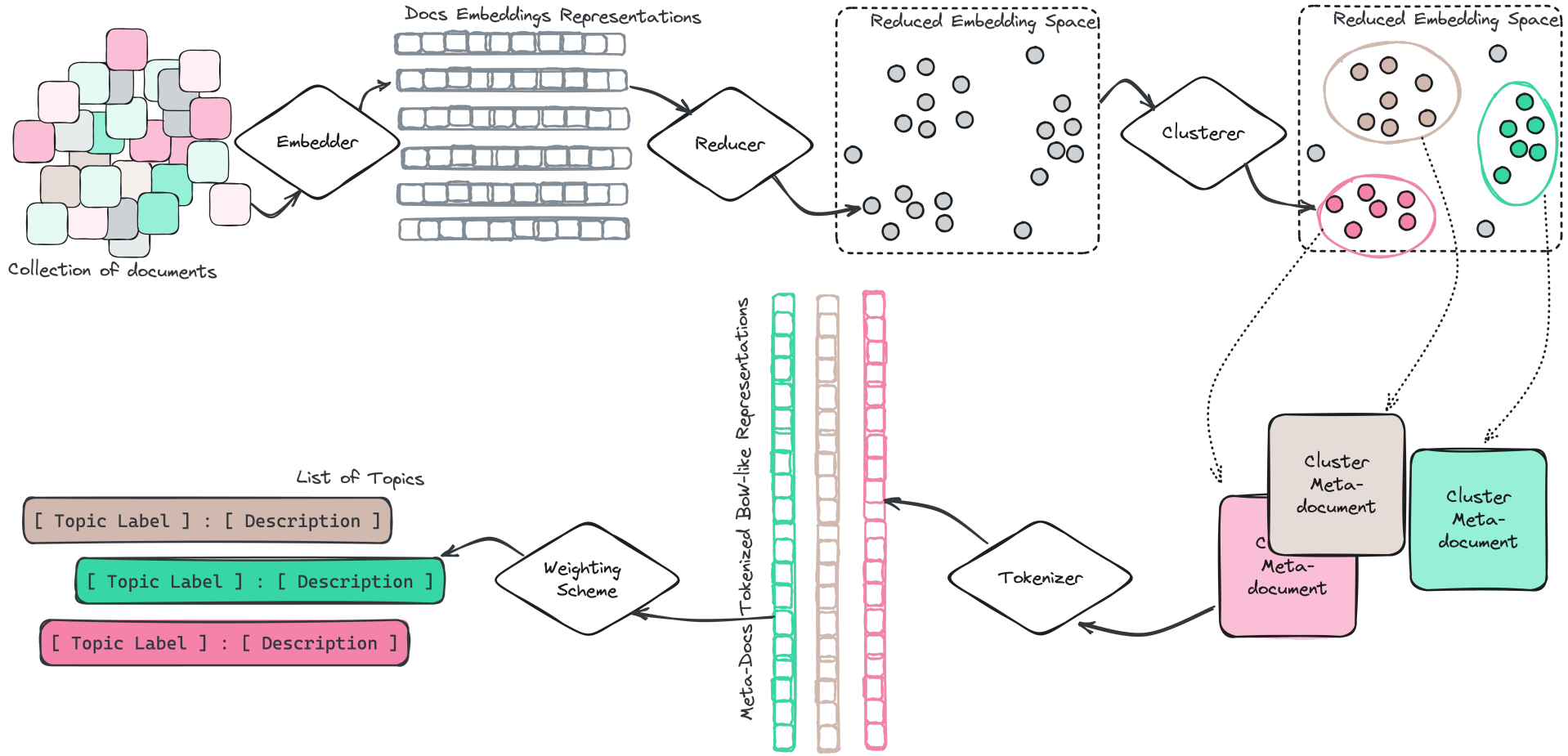
4.2.2 Documents Clustering



4.2.3 Topic Symbolic Representation



4.2.4 BERTopic Autopsy



4.2.5 BERTopic In Practice

```python
from bertopic import BERTopic
texts = [] # your texts here
topic_model = BERTopic()

topics, probs = topic_model.fit_transform(documents)
```

BERTopic from scractch

We'll build a custom implementation of BERTopic during today's Tutorial.

4.3 Latent Dirichlet Allocation (LDA)

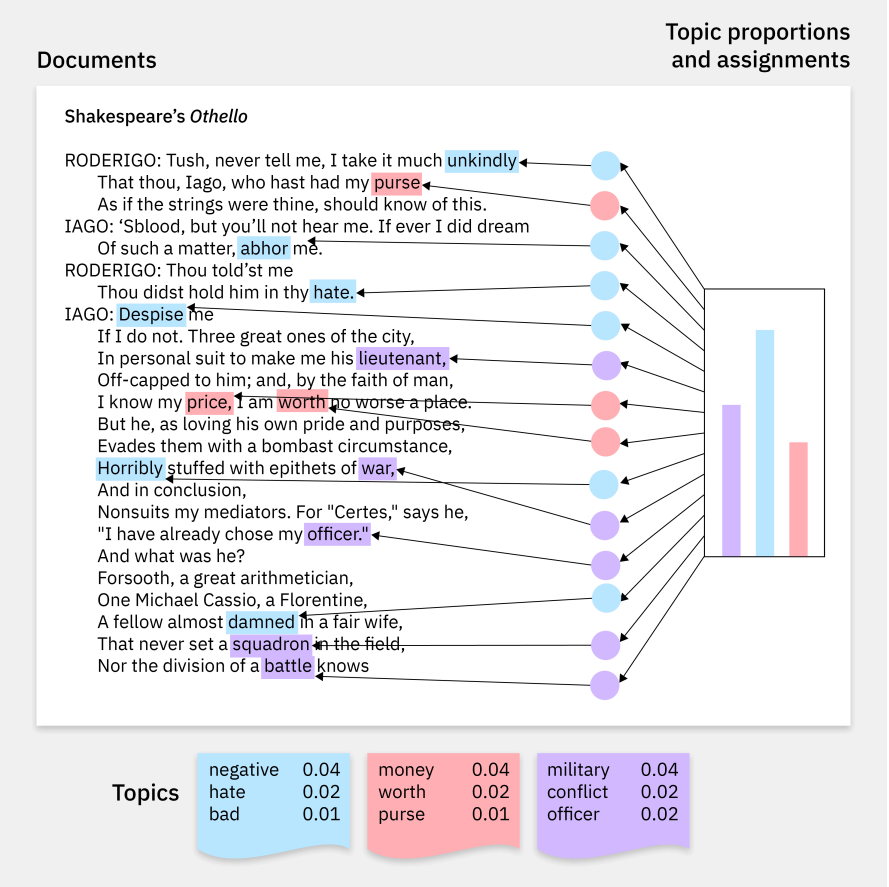
4.3.1 LDA Overview

Latent Dirichlet Allocation (Blei, Ng, and Jordan 2003)
= *proper* TM

LDA Intuition

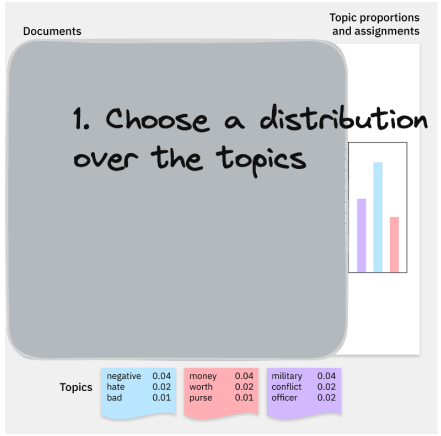
- **Topic** = Probability **distribution** over **words** from fixed vocabulary.
- **Document** = **Mixture** of corpus-wide **topics**.

→ **Generative process**

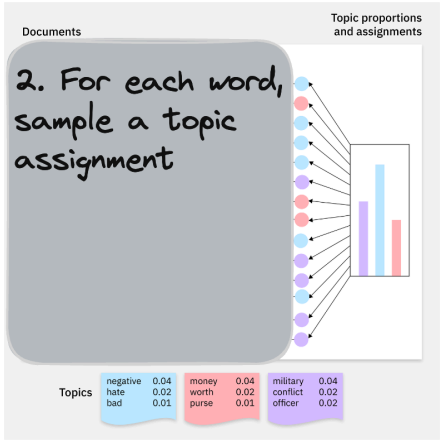


The intuitions behind latent Dirichlet allocation. Extracted from (Murel and Kavlakoglu 2025), inspired by (Blei 2011).

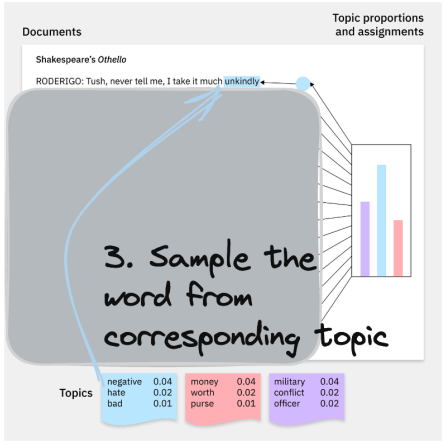
4.3.2 LDA: Generative model



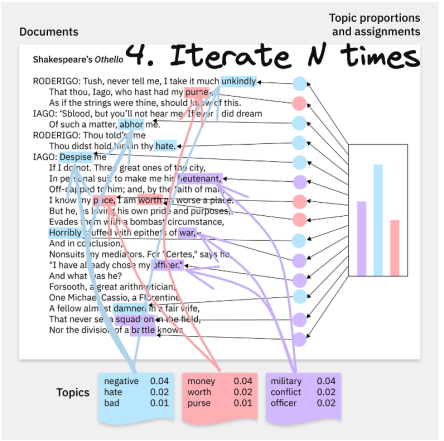
Define doc



> Sample words' topics



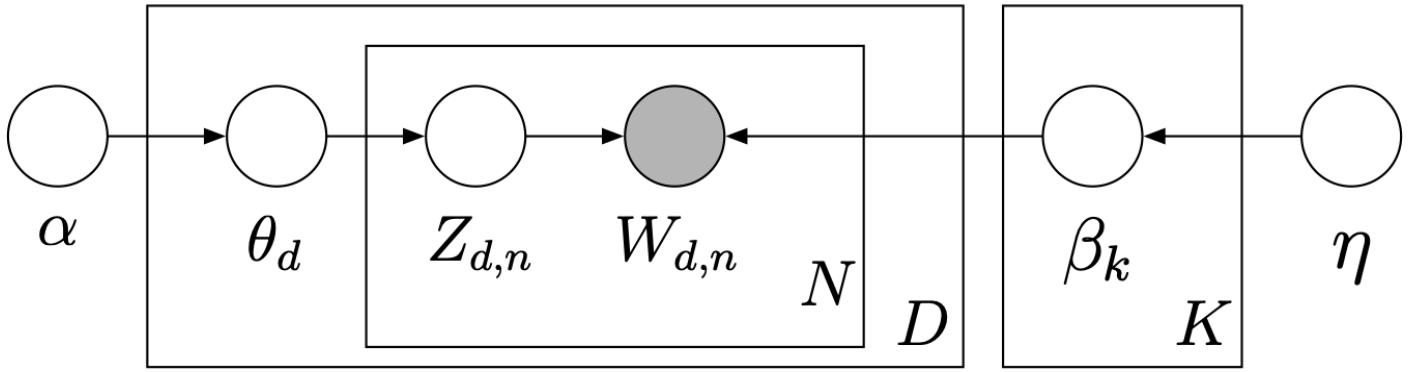
> Sample words from topics



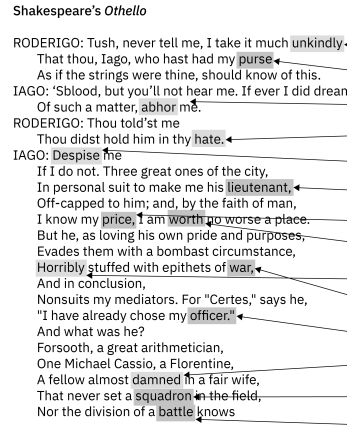
> Iterate

Figure 1: LDA Generative Steps

4.3.3 LDA Graphical Model



LDA Graphical Model. Extracted from (Blei 2011).



Observed State: raw document.

$$p(\beta_{1:K}, \theta_{1:D}, z_{1:D}, w_{1:D}) = \prod_{i=1}^{K} p(\beta_i) \prod_{d=1}^{D} p(\theta_d) \left(\prod_{n=1}^{N} p(z_{d,n} | \theta_d) p(w_{d;n} | \beta_{1:K}, z_{d,n}) \right)$$

4.3.4 LDA In Practice

```python
from gensim.utils import simple_preprocess
from gensim import corpora
from gensim.models import LdaModel

texts = [] # your texts here (might be a good idea to do a bit of cleaning
dictionary = corpora.Dictionary(texts) # extracts vocabulary
corpus = [dictionary.doc2bow(text) for text in texts] # documents BoW repre

lda_model = LdaModel(
    corpus=corpus,
    id2word=dictionary,
    num_topics=92, # YOUR NUMBER OF TOPICS HERE!
    random_state=3,
)
```

4.4 Summary



4.4.1 TM Comparison

	LDA	BERTopic
Approach	Count-based	Neural
# topics	Pre-determined	Automatically determined
Document	Mixture of topics	One topic
Topic	Distribution over words	Extracted from (disjoint) groups of documents

& other approaches (LSA, variants of LDA, NMF, Fastopic, …).

For instance see (Egger and Yu 2022): offers a comparison of diverse TMs applied to Twitter data.

4.5 Evaluation

4.5.1 A hard task!

Reminder: topic models are *unsupervised* (= no ground-truth)

What do we want from a topic model?

Desiderata: *A good TM should…*

1. … ***represent*** *the* ***data*** *well (statistical fit)*
2. … *provide* ***coherent topics*** *(semantic fit)*

→ *How to measure that?*

4.5.1.1 Statistical fit

- Perplexity
  - Assess the ability of a generative process to model observed documents
  - ⚠️ Not applicable to BERTopic!
  - (we'll come back to the notion of *perplexity* in a later class)
- Document-topic similarity
  - Compute average similarity between documents and assigned topic embeddings
  - "Native" for BERTopic, not for LDA (could still be applied)
- Topic Coverage / Diversity
  - Proportion of unique words among the top-N words across topics



Caution

For embedding-based models, *statistical fit* is rarely directly evaluated (rather relies on the evaluation of their components).

4.5.1.2 Semantic fit

Coherence

Topics are good if their top words tend to appear together in similar contexts.

$$\text{Coherence} = \sum_{i<j} \text{score}(w_i, w_j)$$

usually $w_1, \cdots, w_n$ considered are the top n words for a topic by freqeuncy.

Several operationalization of coherence scores:

- UMass (Newman, Noh, et al. 2010): $C_{\text{UMass}} = \log \frac{D(w_i, w_j)+\epsilon}{D(w_i)}$
- UCI (Mimno et al. 2011): $C_{\text{UCI}} = \frac{P(w_i, w_j)+1}{P(w_i)P(w_j)}$
- …

4.5.2 Importance of human judgement

⚠ **Automatic evaluation ≠ human understanding (of topics)**

- Corpus- and parameter-dependent ⇒ to inconsistent comparisons (Newman, Lau, et al. 2010; Röder, Both, and Hinneburg 2015)
- Misalignment with human judgement (J. Chang et al. 2009; Hoyle et al. 2021)
- Optimising metrics can harm interpretability (Mimno et al. 2011)

👀 Human judgement remains essential!

4.6 Topics in Context

4.6.1 Texts Have Context

- Texts do not exist in a void!
- We almost always possess **metadata**:
  - **Author**: Gender, party, demographics
  - **Time**: Date, year, era
  - **Space**: Country, region, GPS coordinate
  - **Genre / Medium**: Speech, tweet, editorial
  - …



Examples of *Topic* + *X* research

- **Literature Themes + Gender**: How does author gender affect the choice and expression of themes in novels? (Jockers and Mimno 2013)
- **Political Debates + Time**: How does Parliament's agenda changes across time? (Greene and Cross 2017)
- **Tweets + Space**: How do expressed topics vary across space? *In London Neighbourhoods (Lansley and Longley 2016); In the US, about environmetal discoures (C. H. Chang, Armsworth, and Masuda 2022)*

→ Beyond Exploration: *How do topics vary across social, demographical, temporal, or geographical dimensions?*



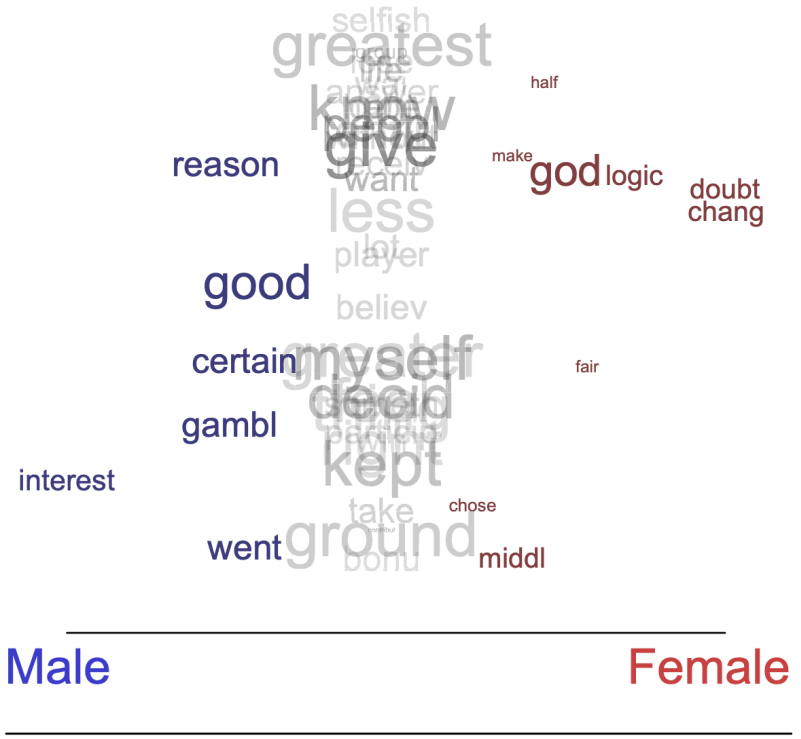

4.6.2 Structural Topic Models (STM)

Generative Integration:

- Standard LDA assumes document-topic distributions (θ_d) come from a fixed prior (α).
- **STM** (Roberts et al. 2013) incorporates metadata directly into the generative process as **covariates**.
  - **Topical Prevalence**: Does document metadata affect *how much* a topic is discussed?
  - **Topical Content**: Does metadata affect *which words* are used to discuss a topic?

→ Uncertainty is propagated end-to-end: topic discovery and covariate estimation happen jointly.

FIGURE 15 Intuitive Topic Allowing for Different Vocabularies Based on Gender



Extracted from (Roberts et al. 2014).

4.6.3 Pointwise Mutual Information (PMI)

Post-Hoc Analysis:

- Fit the model first, then calculate **association metrics** post-hoc.
- **PMI** (Fano and Hawkins 1961):

  How specific is a topic t *to group* m*?*

  - In practice, also Positive PMI (PPMI): $\mathrm{PPMI} = \max(0, \mathrm{PMI})$

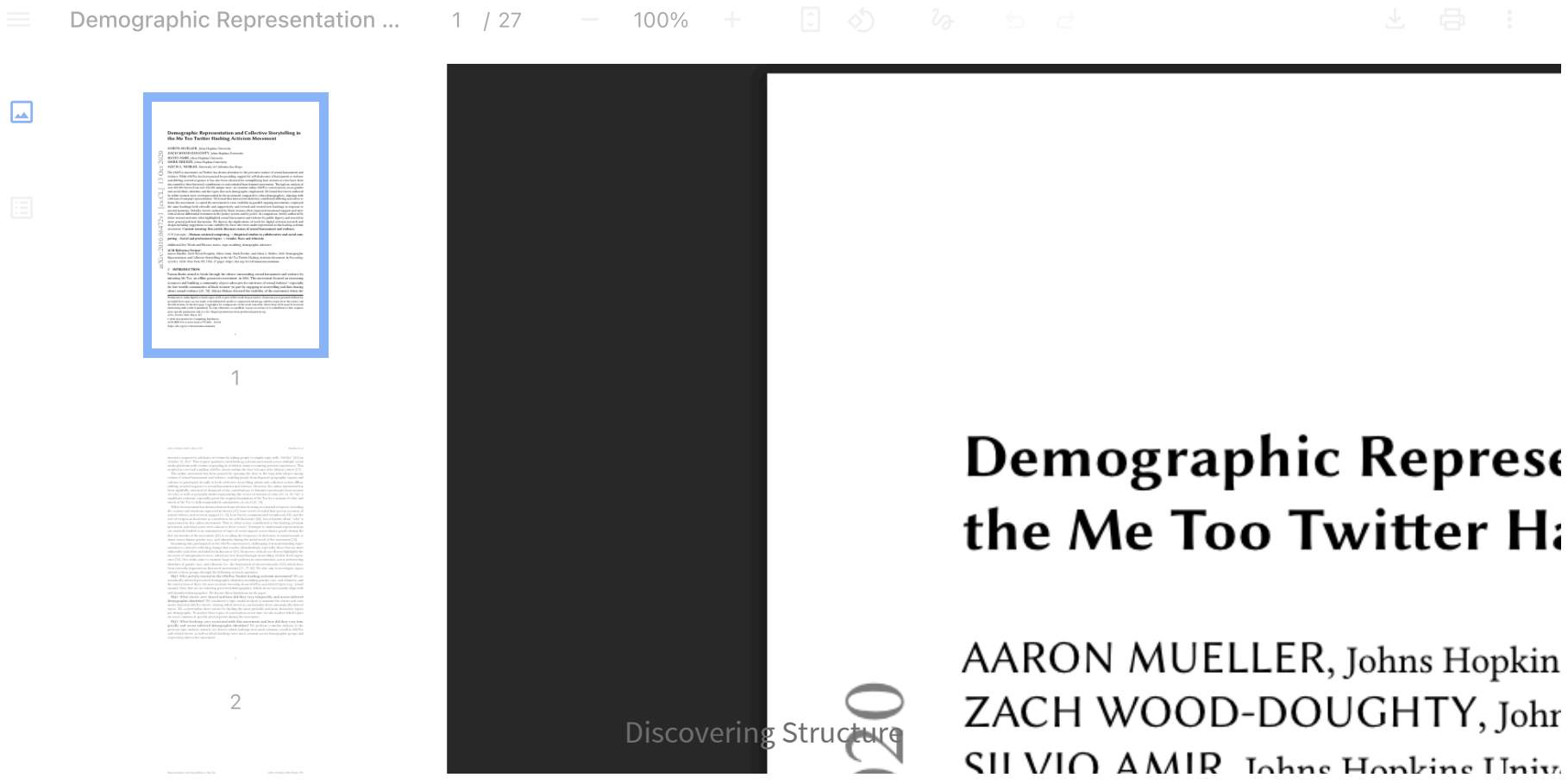
$$\mathrm{PMI}(\mathrm{t}, \mathrm{m}) = \log \frac{\mathrm{P}(\mathrm{t}, \mathrm{m})}{\mathrm{P}(\mathrm{t}) \cdot \mathrm{P}(\mathrm{m})}$$

PMI Terms

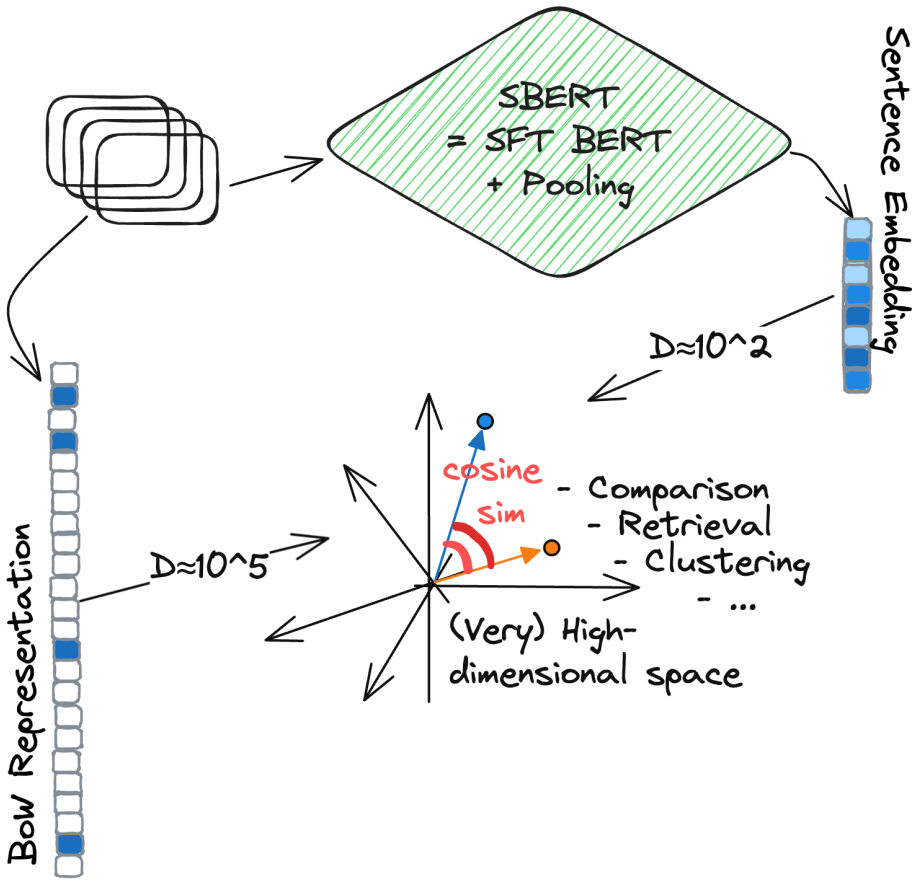
- $\mathrm{P}(\mathrm{t}, \mathrm{m})$: Probability that a document belongs to topic t **and** group m.
- $\mathrm{P}(\mathrm{t})$: Overall prevalence of topic t.
- $\mathrm{P}(\mathrm{m})$: Proportion of documents from group m.

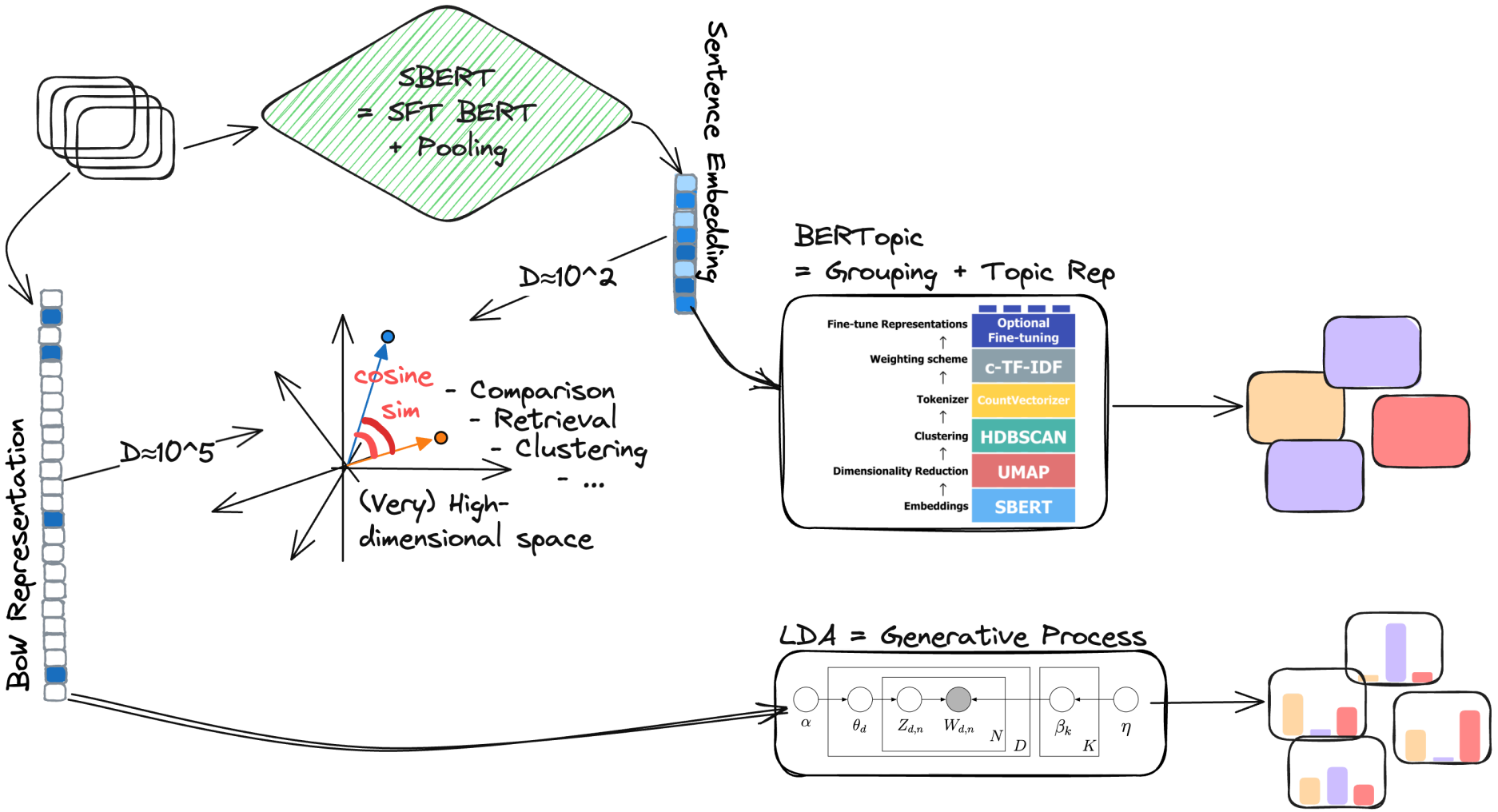
4.6.4 Example

Demographic Representation and Collective Storytelling in the Me Too Twitter Hashtag Activism Movement (Mueller et al. 2021).



5 Takeaways


SBERT
= SFT BERT
+ Pooling
Sentence Embedding
D≈10^2
BoW Representation
D≈10^5
cosine
sim
- Comparison
- Retrieval
- Clustering
- ...
(Very) High-
dimensional space


SBERT
= SFT BERT
+ Pooling
Sentence Embedding
D≈10^2
BERTopic
= Grouping + Topic Rep
Fine-tune Representations
Optional Fine-tuning
Weighting scheme
c-TF-IDF
Tokenizer
CountVectorizer
Clustering
HDBSCAN
Dimensionality Reduction
UMAP
Embeddings
SBERT
Bow Representation
D≈10^5
cosine
sim
- Comparison
- Retrieval
- Clustering
- ...
(Very) High-dimensional space
LDA = Generative Process
α
θd
Zd,n
Wd,n
N
D
βk
K
η

6 Hands-on



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